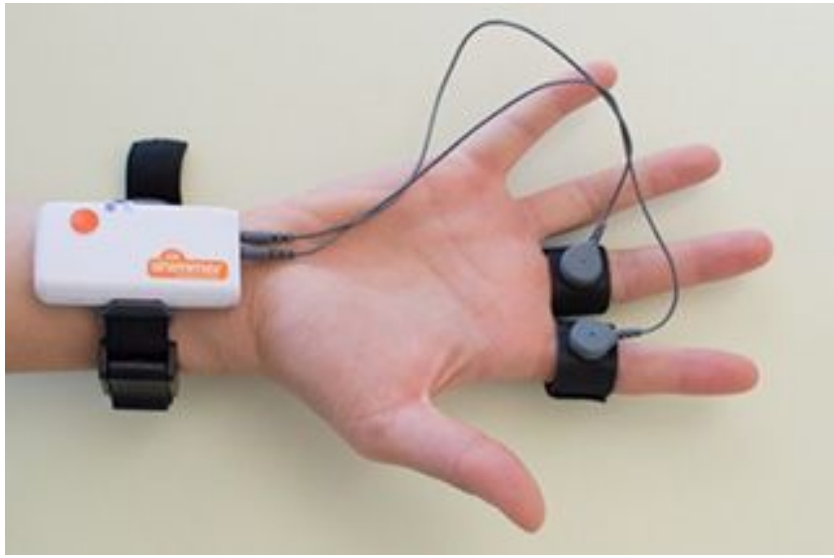
Biomedical Equipments for Clinical Applications | PORTFOLIO

Galvanic Skin Response (GSR) and Basal Skin Response (BSR)



Course 21BMC402J | Unit IV: Sensory Diagnosis and Hearing Aid Equipments | Topic 8: Electrodermal Activity

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Introduction to electrodermal activity

Electrodermal activity (EDA)

- A measure of the average activity of the sweat glands.
- Also a measure of the phasic activity (the high and low points) of these glands.
- Used in biofeedback instrumentation.

BSR

basal skin response

- Electrical resistance of the skin in a subject at rest.
- Gives the baseline value of skin resistance, so it is also called basal skin resistance.

GSR

galvanic skin response

- Change in skin electrical resistance caused by sweat gland activity in the palms and fingers.
- Depends on temperature, humidity and skin moisture level.

Measured most conveniently at the palm, where the body has the highest concentration of sweat glands.

Source: Unit IV notes, Topic 8, pp. 58-61; Khandpur §5.7 (Biofeedback instrumentation), p. 184.

Working principle



Sweat gland activity in palms and fingers → **Skin electrical resistance changes** → **Two electrodes and a dc current source sense the change** → **BSR: baseline | GSR: change**

Skin resistance

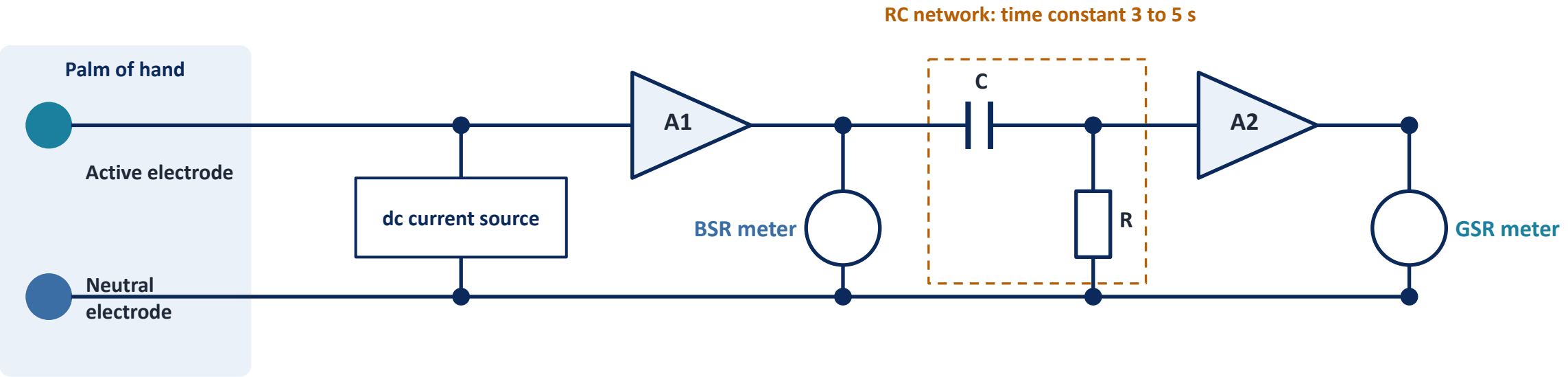
- **GSR:** change in skin resistance caused by sweat gland activity in the palms and fingers.
- **BSR:** resistance of the skin at rest, the baseline value.
- **Dependence:** temperature, humidity and moisture level of the skin.
- **Typical GSR:** 2 to 20 µS, as stated in the notes.

How the electrodes measure it

- **Electrodes:** an active and a neutral electrode placed on the palm.
- **Current:** a dc current source is connected across the electrodes.
- **BSR:** amplifier A1 output is read on the BSR meter.
- **GSR:** an RC network turns the BSR output into a measurable change of skin resistance.

Source: Unit IV notes, Topic 8, pp. 60-62; Khandpur §5.7, p. 184, Fig. 5.18.

Components and block diagram



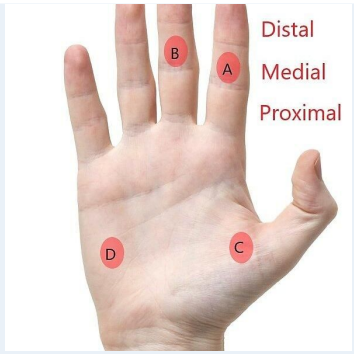
Redrawn after Khandpur Fig. 5.18: block diagram for measurement and record of BSR and GSR (Unit IV notes, p. 62).

- **Electrodes:** active and neutral, on the palm (silver-silver).
- **dc current source:** connected across the electrodes.
- **Amplifier A1:** receives the output; BSR meter reads across its output and ground.
- **RC network:** time constant of 3 to 5 s; lets GSR be measured as a change of skin resistance.
- **A2 and GSR meter:** follow the RC network in Fig. 5.18.

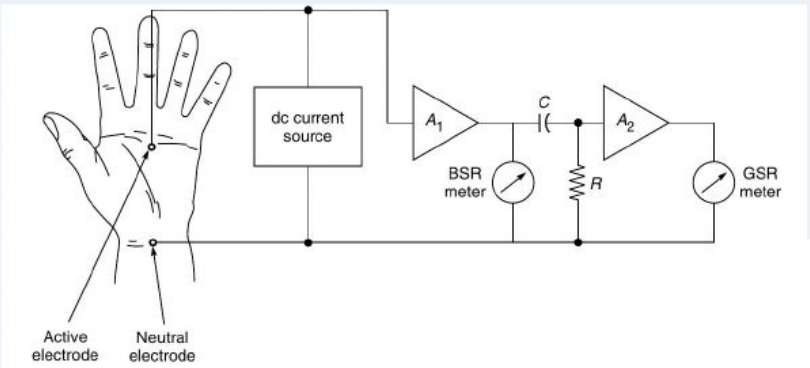
Source: Unit IV notes, Topic 8, pp. 61-62; Khandpur §5.7, p. 184, Fig. 5.18.

Working procedure

1. **Place the electrodes:** Active and neutral electrodes on the palm of the person whose BSR/GSR is to be determined.
2. **Apply dc current:** A dc current source is connected across the two electrodes.
3. **Amplify the signal:** The output is given to amplifier A1.
4. **Measure BSR:** BSR meter across the A1 output and ground gives the baseline (subject at rest).
5. **Record GSR:** BSR output goes to the RC network (3 to 5 s), so GSR is measured as a change of skin resistance.







➤ Fig. 5.18 *Block diagram for measurement and record of Basal Skin Resistance (BSR) and Galvanic Skin Response (GSR)*

Why the palm?
The body has its highest concentration of sweat glands there, so GSR is measured most conveniently at the palm.

Source: Unit IV notes, Topic 8, pp. 61-62; Khandpur §5.7, p. 184, Fig. 5.18.

Applications, advantages and limitations

Applications

- **Biofeedback:** electrodermal activity is used in biofeedback instrumentation.
- **Psychophysiological studies:** such as stress assessment and emotion recognition.
- **Mental disorders:** used in diagnostics and treatment (per the notes).

Advantages

- **Convenient site:** palm, where sweat glands are most concentrated.
- **Simple set-up:** two electrodes, a dc current source, amplifiers and meters.
- **Two readings:** one arrangement gives BSR (baseline) and GSR (change).

Limitations

- **Environment:** depends on temperature, humidity and skin moisture level.
- **Non-specific:** reflects sweat gland activity only.
- **Not diagnostic alone:** GSR cannot by itself diagnose stress or an exact emotion.

Important: GSR alone cannot diagnose stress or determine a person's exact emotion. It shows sweat gland activity and supports, but does not replace, a proper assessment.

Source: Unit IV notes, Topic 8, pp. 60-62. The caution on interpretation is the presenter's statement; advantages are drawn from the described set-up.

Conclusion

- Electrodermal activity reflects sweat gland activity and is measured as BSR and GSR.
- BSR is the baseline skin resistance at rest; GSR is the change in skin resistance.
- Two palm electrodes, a dc current source, amplifier A1 and an RC network (3 to 5 s) make the measurement.
- Used in biofeedback and psychophysiological studies; GSR alone cannot diagnose stress or an exact emotion.

GSR Sensor Demonstration

